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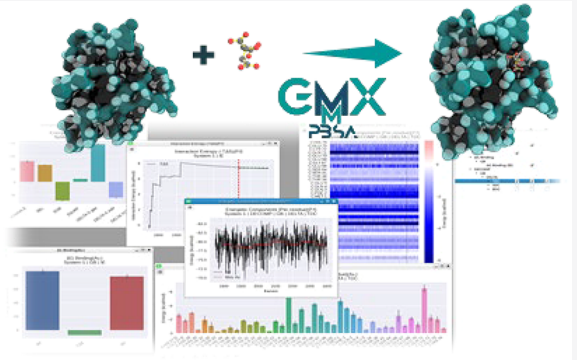
gmx_MMPBSA: A New Tool to Perform End-State Free Energy Calculations with GROMACS

Mario S. Valdés-Tresanco,*[,§] Mario E. Valdés-Tresanco,[§] Pedro A. Valiente, and Ernesto Moreno*

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ABSTRACT: Molecular mechanics/Poisson−Boltzmann (Generalized-Born) surface area is one of the most popular methods to estimate binding free energies. This method has been proven to balance accuracy and computational efficiency, especially when dealing with large systems. As a result of its popularity, several programs have been developed for performing MM/PB(GB)SA calculations within the GROMACS community. These programs, however, present several limitations. Here we present gmx_MMPBSA, a new tool to perform end-state free energy calculations from GROMACS molecular dynamics trajectories. gmx_MMPBSA provides the user with several options, including binding free energy calculations with different solvation models (PB, GB, or 3D-RISM), stability calculations, computational alanine scanning, entropy corrections, and binding free energy decomposition. Noteworthy, several promising methodologies to calculate relative binding free energies such as alanine scanning with variable dielectric constant and interaction entropy have also been implemented in gmx_MMPBSA. Two additional tools—gmx_MMPBSA_test and gmx_MMPBSA_ana—have been integrated within gmx_MMPBSA to improve its usability. Multiple illustrating examples can be accessed through gmx_MMPBSA_test, while gmx_MMPBSA_ana provides fast, easy, and efficient access to different graphics plotted from gmx_MMPBSA output files. The latest version (v1.4.3, 26/05/2021) is available free of charge (documentation, test files, and tutorials included) at https://github.com/Valdes-Tresanco-MS/gmx_MMPBSA.



INTRODUCTION

Molecular mechanics/Poisson−Boltzmann (Generalized-Born) Surface Area (MM/PB(GB)SA) has become one of the most popular methods to estimate binding free energies since its introduction in the late 1990s.[1,2] This method achieves a good balance between accuracy and computational efficiency, being more accurate than most scoring functions, especially when dealing with protein−protein and protein−nucleic acid systems, and less computationally demanding than alchemical free energy methods. MM/PB(GB)SA is a highly versatile method that allows estimating binding free energies, determining structural stability, predicting hotspots, and assessing the contributions from individual residues or energy terms by free energy decomposition analyses.[3−5]

Several computational tools used to perform MM/PB(GB)-SA calculations have been developed and released over the years. MMPBSA.py is a Python script implemented in the Amber package that allows users to quickly and efficiently calculate free energies from molecular dynamics (MD) and Monte Carlo simulations.[6,7] MMPBSA.py has been the leading tool for performing MM/PB(GB)SA calculations within the Amber package due to its improved accessibility, implementation, and capabilities as compared to older scripts like *mmpbsa.pl*.[6] Multiple implicit solvation models are available within MMPBSA.py, including the PB, GB, and 3D-Reference Interaction Site Model (3D-RISM). Normal mode or quasi-harmonic analysis can be used to calculate vibrational frequencies and approximate the solute entropy. Specific interactions can be analyzed using alanine scanning or free energy decomposition. All these procedures can be speeded up by dividing frames evenly across available processors, thus providing the users with an efficient and user-friendly program with the flexibility to accommodate most of their needs.[6,7]

On the other hand, several toolkits have been developed to conduct MM/PBSA calculations on structural ensembles derived from GROMACS[8] trajectories. g_mmpbsa implements the MM/PBSA approach using subroutines written in-house or sourced from the GROMACS and APBS[9] packages. It is distributed as a stand-alone tool with a user interface similar to those of other GROMACS modules. This tool provides a

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choice of nonpolar solvation models and different types of atomic radii to calculate the polar solvation energy. Finally, g_mmpbsa can be used to decompose the total calculated binding energy into contributions per residue.[10]

GMXPBSA is another user-friendly suite of Bash/Perl scripts for streamlining MM/PBSA calculations on GROMACS trajectories.[11] This tool also combines GROMACS and the Poisson−Boltzmann equation solver APBS to dissect the different MM/PBSA energy contributions. GMXPBSA can compare the binding free energy of different complex trajectories, allowing researchers to investigate the effects of alanine and nonalanine mutations, post-translational changes, and non-natural amino acids on the binding free energy of the system under study. A single or distributed automated fashion on a cluster facility can speed up calculations by dividing frames across the available processors.[11] g_mmpbsa and GMXPBSA are distributed for free and used extensively within the GROMACS community. Nevertheless, they have limitations regarding the number of calculations that can be performed and various compatibility issues.

Here, we present a new tool named gmx_MMPBSA designed to fill the gap and bring AmberTools's rich functionalities from MMPBSA.py and associated programs to GROMACS users. Several calculations can be performed within gmx_MMPBSA, including but not limited to binding free energy calculations with different solvation models (PB, GB, or 3D-RISM), stability calculations, computational alanine scanning, entropy corrections, and binding free energy decomposition. Two additional tools have been integrated within gmx_MMPBSA: gmx_MMPBSA_test and gmx_MMPBSA_ana. gmx_MMPBSA_test provides the user with multiple illustrating examples (https://valdes-tresanco-ms.github.io/gmx_MMPBSA/examples) and allows whether gmx_MMPBSA was installed successfully to be checked. gmx_MMPBSA_ana, on the other hand, is a graphical user interface that provides fast, easy, and efficient access to different graphics plotted from gmx_MMPBSA output files. The gmx_MMPBSA suite has been written in Python 3.8 and uses the MMPBSA.py calculation engine, a widely used tool for streamlining end-state free energy calculations. The program is freely available (documentation, test files, and tutorials included) at https://github.com/Valdes-Tresanco-MS/gmx_MMPBSA.

THEORY AND METHODS

The MM/PB(GB)SA method can be used for calculating binding free energies of noncovalently bound complexes (Figure 1).

The free binding energy for a complex can be estimated as follows:

$$\Delta G_{\text{bind}} = \langle G_{\text{COM}} \rangle - \langle G_{\text{REC}} \rangle - \langle G_{\text{LIG}} \rangle \quad (1)$$

where each term to the right in the equation is given by

$$\langle G_x \rangle = \langle E_{\text{MM}} \rangle + \langle G_{\text{sol}} \rangle - \langle TS \rangle \quad (2)$$

In turn, ΔG_{bind} can also be represented as

$$\Delta G_{\text{bind}} = \Delta H - T\Delta S \quad (3)$$

where ΔH corresponds to the enthalpy of binding and $-T\Delta S$ to the conformational entropy after ligand binding. When the entropic term is dismissed, the computed value is the effective free energy, which is usually sufficient for comparing relative binding free energies of related ligands.[5,12]

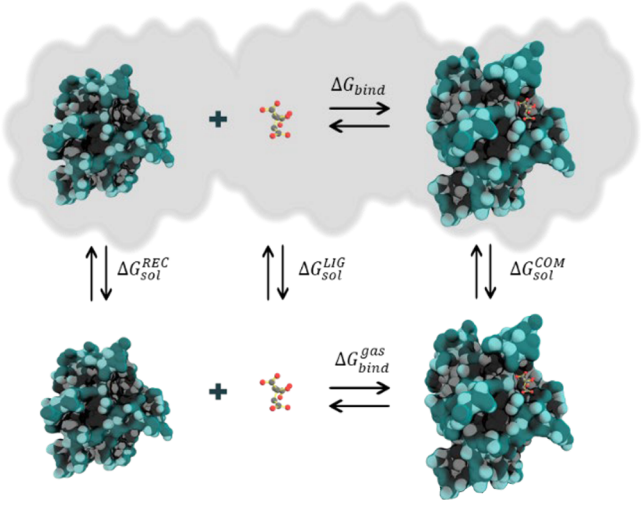


Figure 1. Thermodynamic cycle for binding free energy calculations.

The ΔH can be decomposed into different terms:[1]

$$\Delta H = \Delta E_{\text{MM}} + \Delta G_{\text{sol}} \quad (4)$$

where

$$\Delta E_{\text{MM}} = \Delta E_{\text{bonded}} + \Delta E_{\text{nonbonded}} = (\Delta E_{\text{bond}} + \Delta E_{\text{angle}} + \Delta E_{\text{dihedral}}) + (\Delta E_{\text{ele}} + \Delta E_{\text{vdW}}) \quad (5)$$

and

$$\Delta G_{\text{sol}} = \Delta G_{\text{polar}} + \Delta G_{\text{non-polar}} = \Delta G_{\text{PB/GB}} + \Delta G_{\text{non-polar}} \quad (6)$$

where

$$\Delta G_{\text{non-polar}} = NP_{\text{TENSION}} \times \Delta\text{SASA} + NP_{\text{OFFSET}} \quad (7)$$

or

$$\Delta G_{\text{non-polar}} = \Delta G_{\text{disp}} + \Delta G_{\text{cavity}} = \Delta G_{\text{disp}} + (\text{CAVITY}_{\text{TENSION}} \times \Delta\text{SASA} + \text{CAVITY}_{\text{OFFSET}}) \quad (8)$$

In the above equations, ΔE_{MM} corresponds to the molecular mechanical energy changes in the gas phase. ΔE_{MM} includes ΔE_{bonded}, also known as internal energy, and $\Delta E_{\text{nonbonded}}$, corresponding to the van der Waals and electrostatic contributions. The solvation energy is determined differently, depending on the method employed. In the 3D-RISM model, both components—polar and nonpolar—of the solvation energy are calculated. However, the PB and GB models estimate only the polar component of the solvation. The nonpolar component is usually assumed to be proportional to the molecule's total solvent accessible surface area (SASA), with a proportionality constant derived from experimental solvation energies of small nonpolar molecules (eq 7; Amber's inp1 method).[13,14] Alternatively, a modern approach that separates nonpolar solvation free energies into cavity and dispersion terms can be used (eq 8; Amber's inp2 method). In this approach, SASA is used to correlate the cavity term only, while a surface-integration method is employed to compute the dispersion term.[14]

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Furthermore, the entropic component is usually calculated by normal modes analysis (NMODE). The translational and rotational entropies can be estimated using standard statistical mechanical formulas. Nevertheless, calculating vibrational entropy using normal modes is computationally expensive because it requires expanding the internal coordinate covariance matrix for all degrees of freedom for a set of minimized structures.[15] Conversely, the Quasi-harmonic (QH) approximation is less computationally expensive, although it requires a considerable number of frames to converge.[16] Recently, other alternatives have been developed, such as NMODE in truncated systems, which considerably reduces the computational cost.[17] Interaction Entropy (IE) is another novel method that calculates the entropic component of the binding free energy directly from MD simulations without any extra computational cost. This method is numerically reliable, more computationally efficient, and superior to the standard NMODE approach, as shown in an extensive study of over a dozen randomly selected protein−ligand binding systems.[18]

Typically, two approaches are used for MM/PB(GB)SA calculations, known as Single Trajectory Protocol (STP) and Multiple Trajectory Protocol (MTP).[5,12,19] In STP, both the receptor and the ligand trajectories are extracted from that of the complex. This approach is valid when the bound and unbound states of the receptor and the ligand are similar. It is computationally less expensive than the MTP approach since only a simulation of the complex is required. Additionally, the potential internal terms (e.g., bonds, angles, and dihedrals) cancel exactly in STP since these terms are the same in both bound and unbound states.[5,12] On the other hand, the MTP is a more realistic approach because it considers multiple trajectories (i.e., complex, receptor, and ligand). However, significant conformational changes can lead to numerous errors.[5,12,20] In practice, a detailed study of the system is required to select the approach to be used.

RESULTS AND DISCUSSION

gmx_MMPBSA is written in Python 3.8 and combines functions from GROMACS[8] and AmberTools[12] to generate input files in a consistent/reproducible manner and perform end-state free energy calculations. We will first illustrate the general workflow and how all the tools are integrated into the gmx_MMPBSA environment. Next, we describe the different tools (i.e., gmx_MMPBSA_ana and gmx_MMPBSA_test) that have been built to improve gmx_MMPBSA usability. Later, we include a thorough comparison of gmx_MMPBSA vs other programs widely used within the GROMACS community to perform MM/PBSA calculations. Finally, we present a set of illustrative examples to demonstrate the versatility and potential applications of gmx_MMPBSA.

Workflow. gmx_MMPBSA divides work into three major steps: (i) preparation, (ii) calculation, and (iii) visualization and analysis (Figure 2).

i. Preparation. The MMPSA.py calculation engine is used within gmx_MMPBSA to perform most of the calculations. The preparation process is focused on generating the topologies in Amber format in a consistent and reproducible way. In this step, the PDB structure files for the receptor and ligand are generated from the GROMACS structure+mass(db) files (i.e., *.tpr, *.pdb, or *.gro). Of note, GROMACS differs in several aspects from Amber in terms of data structure. Amber renumbers residues starting from 1, while GROMACS is more flexible, allowing the same residue numbers in different chains

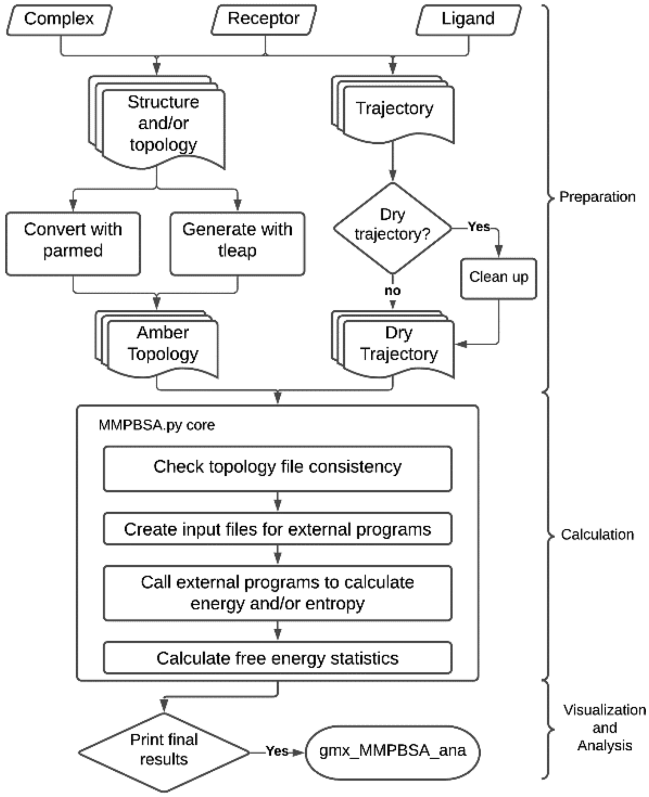


Figure 2. gmx_MMPBSA general workflow for performing end-state calculations. Step 1. Preparation: parmed/LEaP is used to generate Amber topologies, and GROMACS is used to generate PDB structures and clean up the trajectories. Step 2. Calculation: MMPBSA.py core with modifications is used to perform the energy calculations. Step 3. Visualization and analysis: gmx_MMPBSA_ana is used to visualize and analyze the results.

and even residues with insertion codes. Depending on the defined GROMACS structure+mass(db) file, the PDB structure files generated for the receptor and ligand may contain ambiguities or lack a correct notation, which is resolved automatically within gmx_MMPBSA. Another way to overcome these issues within gmx_MMPBSA is by defining a reference structure that can be used to ensure an exact match between the GROMACS structure+mass(db) file and the generated PDB structure files.

Both AMBER and CHARMM force fields are supported, although the input files are created differently (Figure S1). The topologies for AMBER, as a native force field within AmberTools, can be generated either from PDB structures or by converting the GROMACS topology to Amber format. On the other hand, the topologies for the CHARMM force field can be generated only by converting the GROMACS topology to Amber format. The PDB structures generated for the receptor and ligand are used to create the topologies from structures by using LEaP[7] or splitting the GROMACS topologies with parmed after their conversion to Amber format.[21]

Either the input PDB structure for LEaP or the input topology for parmed are thoroughly reviewed. In the PDB structures, the names of the different atoms (i.e., CD carbons in isoleucine and C-terminal oxygens) and residues (disulfide cysteines, histidines, lysines, aspartates, and glutamates) are changed according to the AMBER force field naming convention. The chain IDs are fixed, and if required, a

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Table 1. Comparison of Calculation Features among g_mmpbsa, GMXPBSA2.1, MMPBSA.py, and gmx_MMPBSA

feature	g_mmpbsa	GMXPBSA2.1	MMPBSA.py	gmx_MMPBSA
effective binding free energy	PB	PB	GB, PB, 3D-RISM	GB, PB, 3D-RISM
stability			★	★
GB models			1, 2, 5, 7 and 8	1, 2, 5, 7 and 8
QM-MM/GBSA			★	★
membrane protein			★	★
entropy corrections			NMODE, QH	NMODE, QH, IE
decomposition schemes	per-residue		per-residue, per-wise	per-residue, per-wise
alanine scanning	★[a]	★	★	★
approaches	STP	STP, MTP	STP, MTP	STP, MTP

[a]Without documentation.

mutation is performed for alanine scanning. Additionally, waters and salt ions are removed. In essence, the output of the preparation step is a pair of *.prmtop and *.inpcrd files generated for the structures in the gas phase, which can be used as input files for the MMPBSA.py calculation engine.

MMPBSA.py is one of the most popular and complete software packages for performing MM/PB(GB)SA calculations. This tool is released with AmberTools, and as such, its engine requires Amber *.prmtop topology files as input. Several tutorials describe how to generate Amber topologies from GROMACS files; however, they are cumbersome and multistep processes, leading to errors. Within gmx_MMPBSA, we have optimized this process to handle large and heterogeneous systems. Additionally, we have included several Amber-compatible force fields (i.e., zwitterionic amino acids[22]) to ensure compatibility with as many systems as possible. A comprehensive list of the different systems tested so far can be found at https://valdes-tresanco-ms.github.io/gmx_MMPBSA/examples.

Unlike MMPBSA.py, which uses an additional tool (ante-MMPBSA.py) to generate the topologies for the STP approximation and change the atomic radii, gmx_MMPBSA performs the process internally. gmx_MMPBSA separates the receptor and ligand in the preparation step, avoiding the requirement for additional procedures or external applications. Similarly, our tool generates the alanine mutant without an external tool or a user-defined topology. This reduces unforced errors and speeds up the input file preparation process.

ii. Calculation. Multiple calculations (e.g., binding free energy with different solvation models (PB, GB, or 3D-RISM), stability, alanine scanning, entropy corrections, and binding free energy decomposition) can be conducted with gmx_MMPBSA. The preparation and execution of alanine scanning and energy decomposition have been substantially facilitated as compared to their implementation in MMPBSA.py. As a new facility, interaction entropy calculations can now be conducted. For these new features, several variables in the input file have been added or modified in gmx_MMPBSA. A comprehensive description of the added/modified variables can be found at https://valdes-tresanco-ms.github.io/gmx_MMPBSA/input_file.

iii. Visualization and Analysis of Results. Once the calculation is finished, a graphical user interface application (gmx_MMPBSA_ana) shows up to visualize and analyze the results. A detailed description of gmx_MMPBSA_ana functionalities is provided below.

gmx_MMPBSA Tools. gmx_MMPBSA includes three tools: gmx_MMPBSA, which constitutes the application core; gmx_MMPBSA_ana, a tool for visualizing and analyzing the results; and gmx_MMPBSA_test, which helps users to familiarize themselves with the application, as well as inspect a set of provided examples.

gmx_MMPBSA. gmx_MMPBSA carries out the first two steps (i.e., preparation and calculation) described in the workflow. Additionally, it has an application programming interface (API) that provides direct access to the raw data produced during the calculation (https://valdes-tresanco-ms.github.io/gmx_MMPBSA/summary/#gmx_mmpbsa-in-a-nutshell).

gmx_MMPBSA_ana. gmx_MMPBSA_ana is a graphical user interface (GUI) tool that has been built with the Python PyQt5 library for analyzing gmx_MMPBSA results. This tool provides a clean and user-friendly work environment using a multidocument interface that eases the graph analysis process, either individually or in sets. gmx_MMPBSA_ana has a multiprocessing capacity for reading and loading the results, thus reducing the processing time considerably. It uses seaborn[23] and matplotlib[24] for plotting, as well as pandas[25] and numpy[26] for manipulating the data. Currently, gmx_MMPBSA_ana allows analyzing multiple systems simultaneously and computing the correlation between calculated and experimental energies in a user-defined set. Several graphs can be plotted in gmx_MMPBSA_ana, for instance, per-frame line and average bar graphs for each energy component, temporal heatmaps for per-residue free energies, relational heatmaps for per-wise free energies, and linear regression graphs for correlation analyses between the systems. Of note, each chart has a toolbar that allows editing several features in the graphs and exporting high-quality images. A thorough review of all kinds of graphs supported in gmx_MMPBSA_ana can be found at https://valdes-tresanco-ms.github.io/gmx_MMPBSA/analyzer.

gmx_MMPBSA_test. We have currently provided gmx_MMPBSA users with a set of examples that show how the program works and the variety of systems and analyses that gmx_MMPBSA supports. Users can access them directly from the repository or by using gmx_MMPBSA_test. This tool downloads the current version of the example folder from the repository and uses exactly the command line recommended in each example. The process is carried out using multiprocessing, which allows running the examples in less time. Once all the calculations are finished, gmx_MMPBSA_ana is launched to display the results of all the selected systems. In this way, we delivered a tool to test gmx_MMPBSA operation and familiarize the user with the characteristics, methods, and analyses available in our application (https://valdes-tresanco-ms.github.io/gmx_MMPBSA/command-line/#running-gmx_mmpbsa_test).

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Table 2. Comparison of Analysis Features among g_mmpbsa, GMXPBSA2.1, MMPBSA.py, and gmx_MMPBSA

feature	g_mmpbsa	GMXPBSA2.1	MMPBSA.py	gmx_MMPBSA
API			★	★
analyzer tool			*a*	gmx_MMPBSA_ana
multiple systems at the same time				★
correlation between systems	×			★
per-residue energies to PDB	×			★
interactive visualization				★
3D molecular visualization				PyMOL
interactive charts	static image			★
plotting tool	internal tools		API and graphics library[b]	gmx_MMPBSA_ana
energetic term charts	G_{polar}, $G_{nonpolar}$, E_{MM}, G_{bind}			all
export data to CSV file			★	★
energy summary			★	★
individual energetic terms				★

[a]We plan to extend gmx_MMPBSA_ana compatibility to MMPBSA.py's results. [b]Currently, there is a repository (https://github.com/williamdlees/AmberUtils) for analyzing the results.

gmx_MMPBSA vs Other Programs. Hereunder, we compare gmx_MMPBSA with the different existing GROMACS programs to perform MM/PBSA calculations. The comparison includes g_mmpbsa,[10] a tool widely used within the GROMACS community, and GMXPBSA2.1.[11] We have also considered MMPBSA.py in the comparison, as it constitutes the base for gmx_MMPBSA.

Calculation Features. gmx_MMPBSA includes the GB, PB, and 3D-RISM models for estimating the solvation energy during binding free energy or stability calculations. Thus far, no program allows GROMACS users to perform calculations using different GB models. The previously licensed Sander code was recently released, thus enabling unrestricted use of the GB^{HCT},[27] GB^{OBC}, GB^{OBC2},[28] GB-Neck,[29] and GB-Neck2 models.[30] These models are widely used in free energy calculations as they allow achieving good results at a reasonable computational speed. In addition, part of the system can be treated by quantum mechanics using a hybrid quantum mechanical−molecular mechanical (QM/MM) Hamiltonian and a GB model, thus allowing a more accurate description of the system (Table 1). The model given by Pellegrini and Field[31] is used in the QM/MM generalized Born implementation, with QM/MM PME support based on the work by Walker et al.[32]

Furthermore, several finite-difference numerical solvers are available through gmx_MMPBSA, such as linear PB solvers,[33,34] which can be used for standard applications such as modeling of globular proteins.[35] On the other hand, nonlinear PB solvers[36] are recommended when dealing with highly charged systems, such as nucleic acids. An implicit membrane region with a uniform[37] or heterogeneous[38] dielectric constant can also be considered in the calculations (Table 1).[39] Neither g_mmpbsa nor GMXPBSA2.1 has this feature implemented despite the APBS capability to simulate a membrane environment.[40]

gmx_MMPBSA also incorporates 3D-RISM for calculating solvation energies (Table 1). 3D-RISM is based on first-principles and gives a comprehensive molecular description of the solvation.[41,42] It considers various interactions between the solvent and the solute (e.g., hydrophobic interactions, hydrogen bonds), making it the most rigorous statistical mechanics solvation model available in gmx_MMPBSA. Nevertheless, it is prohibitive for several applications due to the large memory and CPU time required for the calculations.

Furthermore, gmx_MMPBSA implements a new method for estimating the entropic term IE method, in addition to the two (NMODE and QH) already available in MMPBSA.py. The main advantage of this new method is its low computational cost, making it attractive for high throughput energy calculations.[43−45] Neither g_mmpbsa nor GMXPBSA2.1 has any of these methods implemented in their routines. Although several studies conclude that correlations based on effective energy may be sufficient,[3,46] others assert that entropic correction can improve these estimates.[43,47,48] In any case, we provide the user with different methods to estimate the entropic component that can be executed independently (Table 1).

Although the GB and PB models are not strictly decomposable, the per-wise scheme can provide information about pairwise interactions between residues. Similarly, both alanine scanning and per-residue decomposition allow estimating the contribution of a particular selection of residues. Currently, g_mmpbsa only supports residue decomposition, while GMXPBSA2.1 only supports alanine mutation at several residues simultaneously. These analyses shed light on the receptor−ligand interaction, though the results of a per-residue or per-wise decomposition should be interpreted carefully. gmx_MMPBSA implements a more flexible selection method than MMPBSA.py when selecting residues from the complex to print. The interacting elements in the receptor and the ligand within a certain distance can be spotted by using the "within" keyword in gmx_MMPBSA (see an example at https://valdes-tresanco-ms.github.io/gmx_MMPBSA/examples/Decomposition_analysis). This reduces considerably both the computation time and the amount of generated raw data, as well as errors when listing the residues (Table 1).

Finally, gmx_MMPBSA implements both the STP and MTP approach. The user can decide which approach to use based on the system's behavior during the production run. Only GMXPBSA2.1 allows comparison between different systems (e.g., comparing a mutant and the wild type), which will be implemented soon in gmx_MMPBSA. Additionally, gmx_MMPBSA allows defining several trajectories of the same system, allowing the analysis of multiple replicas in the same calculation without concatenating them previously (Table 1).

Analysis Features. gmx_MMPBSA is the only program that includes a dedicated analysis tool among all the compared

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Table 3. Comparison of Technical Features among g_mmpbsa, GMXPBSA2.1, MMPBSA.py, and gmx_MMPBSA

feature	g_mmpbsa	GMXPBSA2.1	MMPBSA.py	gmx_MMPBSA
GROMACS	4.x.x, 5.x.x, and 2016+[a]	4.x.x, 5.x.x, and 20xx.x[b]		4.x.x, 5.x.x, and 20xx.x
dependencies	APBS (1.2.x, 1.3.x or 1.4.x)[c]	APBS (1.x.x)[d,e]		AmberTools20/21[e]
parallel computation	depends on APBS	★	★	★
steps for:				
calculation and summary	multiple	multiple	one	one
analysis	multiple	multiple	multiple	one

[a]GROMACS 20xx.x is not officially supported. There is a Pull Request that offers minimum compatibility with versions higher than 2016.x, but still with limitations. [b]GROMACS 20xx.x is not officially supported. It requires several modifications of the scripts to work correctly. [c]The precompiled program does not require any external library or GROMACS and APBS package. [d]It is not clear whether it supports APBS versions 3.x.x. [e]The referred program must be installed.

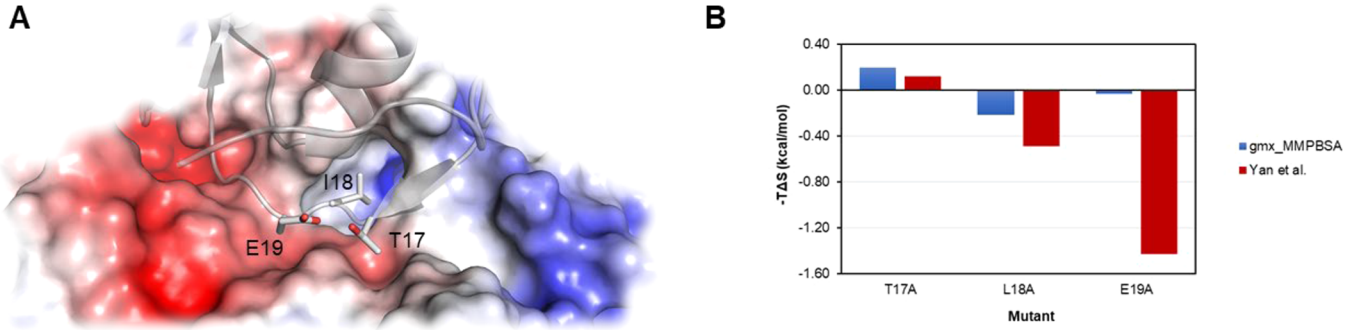


Figure 3. Relative IE values for OMTKY3 mutants. (A) Chymotrypsin:OMTKY3 complex. Residues in the OMTKY3 reactive site loop are represented as sticks. The electrostatic surface is shown for chymotrypsin. (B) Comparison of IE relative values obtained with gmx_MMPBSA vs those reported by Yan et al.

programs. gmx_MMPBSA_ana provides a friendly GUI that allows access to all the gmx_MMPBSA results. Multiple systems can be analyzed within gmx_MMPBSA_ana concurrently. Pearson and Spearman's coefficients can be calculated with their associated p-values, and scatter plots containing the calculated vs experimental energy are provided. Noteworthy, up to 12 different methods (i.e., PB, GB, 3D-RISM, PB+NMODE, PB+QH, PB+IE, GB+NMODE, GB+QH, GB+IE, 3D-RISM+NMODE, 3D-RISM+QH, and 3D-RISM+IE) can be used to calculate energies in the same gmx_MMPBSA run, given that PB/GB/3D-RISM and entropic calculations are performed independently. This way, users can select the best method for their system in a single step (Table 2).

On the other hand, line plots show instantaneous energy values for individual (i.e., E_{ele}, E_{vdW}) or summed terms (i.e., G_{gas}, G_{sol}, G_{total}). Likewise, bar plots are used to show average values for different terms with their respective standard deviations. In addition, heatmap plots are used to show results from per-residue and per-wise energy decomposition (Table 2).

Additionally, gmx_MMPBSA_ana incorporates the interactive visualization of per-residue energy contributions on the PDB structure using PyMOL.[49] PyMOL is widely used to visualize, share, and analyze molecular data. In addition to being lightweight and fast, PyMOL can create images of peerless visualization quality while offering the flexibility of Python-based development and scalability.[49] Nonetheless, the PDB file with the mapped per-residue energy contributions can also be exported and visualized with other software (e.g., Chimera or VMD, see https://valdes-tresanco-ms.github.io/gmx_MMPBSA/examples/Decomposition_analysis). Finally, the raw data can be exported in CSV format for downstream analyses (Table 2).

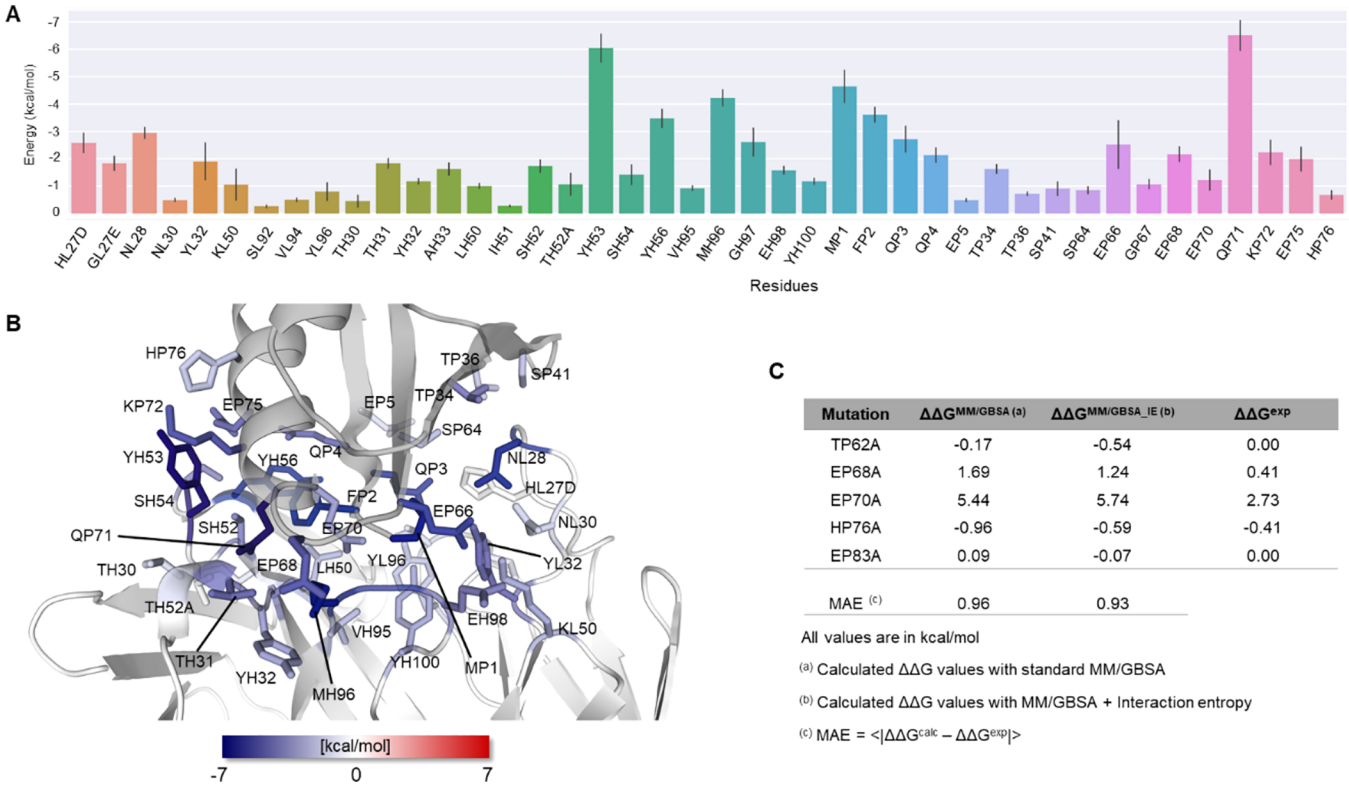
Technical Features. g_mmpbsa and GMXPBSA2.1 present some compatibility issues with modern GROMACS versions (20xx.x), while gmx_MMPBSA can use any GROMACS version included in the PATH to extract the structures, generate the indexes, and clean up the trajectories. We have successfully tested several GROMACS versions, including but not limited to 4.6.5, 5.1.2, 2018.3, 2019.1, 2020.3, and 2021.1 (Table 3).

All the programs can run in parallel, although there are differences in their execution and implementation. g_mmpbsa depends exclusively on the APBS parallelization; thus, it is necessary to configure both the external use of APBS and environment variables that control the number of processors to use. Conversely, GMXPBSA2.1 takes advantage of both locally available processors and PBS/Torque queuing in a cluster. Both gmx_MMPBSA and MMPBSA.py natively incorporate parallelization with Message Passing Interface (MPI) through the mpi4py library.[50] That way, it efficiently leverages either processors available locally or on multiple nodes in high-performance computing (HPC) systems. In addition, it does not require any additional process, execution of external applications, or user intervention (Table 3).

The calculation and analyses are performed as a single-step execution process within gmx_MMPBSA. On the contrary, several python scripts are required in g_mmpbsa to perform the final statistical analyses and calculate the energetic contributions of each residue in the complex. Likewise, GMXPBSA2.1 comprises a suite of Bash/Perl scripts for streamlining MM/PBSA calculations (Table 3).

Illustrating Examples. To demonstrate the applicability and versatility of the gmx_MMPBSA module, we performed

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Mutation	ΔΔG$^{MM/GBSA\ (a)}$	ΔΔG$^{MM/GBSA_IE\ (b)}$	ΔΔG^{exp}
TP62A	-0.17	-0.54	0.00
EP68A	1.69	1.24	0.41
EP70A	5.44	5.74	2.73
HP76A	-0.96	-0.59	-0.41
EP83A	0.09	-0.07	0.00
MAE [c]	0.96	0.93	

All values are in kcal/mol

[a] Calculated ΔΔG values with standard MM/GBSA

[b] Calculated ΔΔG values with MM/GBSA + Interaction entropy

[c] MAE = <|ΔΔG^{calc} − ΔΔG^{exp}|>

Figure 4. MMGBSA calculations for jel42 Fab fragment/HPr complex. (A) Per-residue energy contributions to the formation of the jel42 Fab fragment/HPr complex. (B) Structural representation of residues at the interface of the complex colored according to their energy contribution to the complex formation; see color-gradient bar. (C) Calculated and experimental ΔΔ*G* values for HPr mutants.

several calculations for complexes of different nature, i.e., protein−inhibitor, antibody−antigen, and protein−carbohydrate. To check the correct implementation of gmx_MMPBSA, we ran these systems also with MMPBSA.py, using the topologies generated with gmx_MMPBSA and an input file with the same variables, as well as the trajectories as inputs to MMPBSA.py. In all cases, the results were identical. Details on system preparation, MD simulation procedures, and energy calculations are described in the Supporting Information.

gmx_MMPBSA Reproduces the Relative IE Values of Alanine Mutants. Yan and co-workers[43] proposed the IE method combined with residue-specific dielectric constants to compute the entropic contribution in alanine scanning calculations. They validated this method in a set of 24 protein−protein complexes. An overall mean absolute error of 1.0 kcal/mol relative to the experiment was achieved when combining MM/GBSA, IE, and residue-specific dielectric constants (i.e., 1 for nonpolar residues, 3 for polar residues, and 5 for charged residues).[43] Here, we chose a well-studied system, chymotrypsin, with its inhibitor turkey ovomucoid third domain (OMTKY3), to test the IE method implementation in gmx_MMPBSA.

The relative IE values of three alanine mutants (T17A, I18A, and E19A) in the reactive site loop of OMTKY3 (Figure 3A) were calculated with gmx_MMPBSA, showing a good agreement with those reported by Yan et al. (Figure 3B). The calculated values are modularly small, which is expected since none of these residues changes their conformation significantly upon complex formation.[51] The most significant difference was observed for E19, a charged residue exposed to the solvent. As noticed recently,[52] the IE values for charged residues are more difficult to converge and reproduce because their interaction energies tend to be higher and more variable. It is worth noting also that the study by Yan et al. was carried out with Amber, while our simulations were conducted with GROMACS. Even if the same parameters are used to generate the trajectories, this does not ensure that the evolution of the system will be the same. Therefore, these factors may account also for the obtained differences.

Energy Contributions of Residues at the Antibody−Antigen Complex Interface. Jel42 Fab fragment/HPr (PDB ID: 2JEL)[53] is an antibody−antigen complex determined at 2.5 Å resolution. Experimental information from alanine-scanning mutagenesis experiments is accessible through the SKEMPI database[54] and was used to validate our findings. The antibody residues are numbered according to Kabat numbering system.[53,55,56] H, L, or P are the chain identifiers for the heavy and light chains of the Jel42 Fab fragment and the HPr antigen, respectively.

The identification of hot spots at the interface was first carried out using the per-residue effective free energy decomposition protocol. This method predicted a high free energy contribution ($|\Delta G_{res}| > 6$ kcal/mol) for QP71 and YH53 (Figure 4A). QP71 is buried in the antibody binding site and establishes multiple van der Waals contacts, hydrogen bonds with TH30 and TH52A, and water-mediated interactions with SH52 and SH54 residues.[53] YH53 presents a similar interaction profile, interacting through hydrogen bonding with HP76 and van der Waals contacts with QP71, KP72, and EP75 (Figure 4B).[53] Residues YH56, MH96, MP1, and FP2 are predicted to have a high ($|\Delta G_{res}| > 3$ kcal/mol) contribution to the effective binding free energy (Figure 4A). These residues interact mainly through van der Waals contacts

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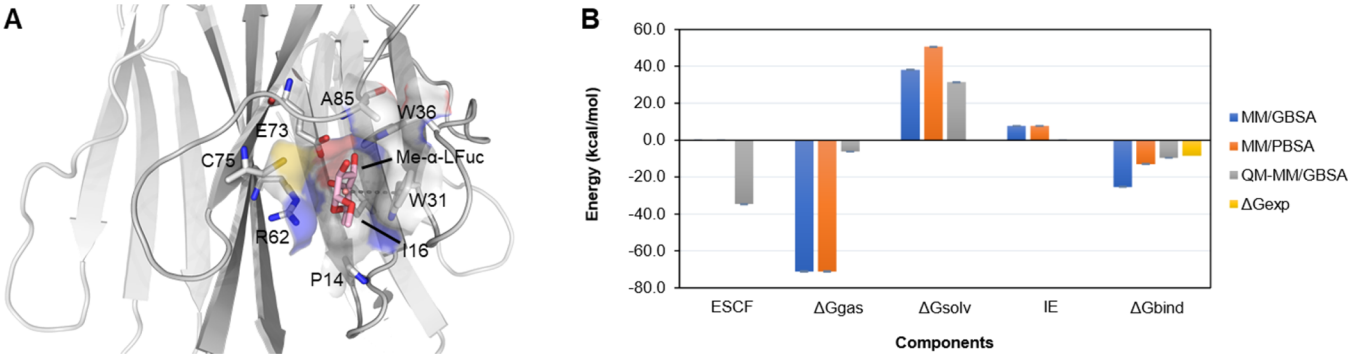


Figure 5. Assessment of different methods to calculate the binding free energy of the Lectin RSL:Me-α-L-Fucose complex. (A) Me-α-L-Fucose bound to the intermonomeric site of Lectin RSL. CH$\cdots\pi$ stacking is shown as gray dashed lines. (B) Performance of MM/GBSA + IE and MM/PBSA + IE vs QM-MM/GBSA. Error bars represent standard errors from a single run.

favored by their large hydrophobic side chains (Figure 4B).[53] Of note, the contribution to the free energy of the residues mentioned above was very stable throughout the MD simulation (Figure S2).

Next, computational alanine scanning was performed over five residues (TP62, EP68, EP70, HP76, and EP83). The results of standard MM/GBSA calculations and those including the entropic contribution computed with the IE method were compared directly to the experimental data. As shown in Table 2C, both approaches show encouraging results with mean absolute errors (MAE) < 1 kcal/mol, which denotes excellent agreement with experimental findings. A detailed analysis of the contributions showed the enthalpic component (ΔH) as favorable (negative value) to the binding process. At the same time, the entropic term ($-T\Delta S$) was unfavorable (positive value) and modularly smaller (Figure S3).

Comparing QM-MM/GBSA vs MM/PB(GB)SA + IE Method. Lectin RSL from *Ralstonia solanacearum* complexed with Me-α-L-Fucose was obtained from the X-ray crystal structure (PDB ID: 2BT9)[57] and used as a study case to evaluate the performance of QM-MM/GBSA vs MM/PB(GB)SA + IE method.

Figure 5A shows Me-α-L-Fucose bound to the intermonomeric site of Lectin RSL. Noteworthy, W31 establishes a CH$\cdots\pi$ stacking interaction with Me-α-L-Fucose.[57] This interaction between an aromatic amino acid and the hydrophobic face of the carbohydrate can contribute up to 5 kcal/mol.[58] Figure 5B shows a comparison among different methods to calculate the binding free energy. In this particular case, the PB model performed better than the GB model. Furthermore, the addition of the entropic component improves the calculations (ΔG_{calc} = −12.7 ± 2.5 kcal/mol with PB; ΔG_{calc} = −25.2 ± 2.1 kcal/mol with GB), although the binding free energy value calculated with the GB model is still far from the experimental one (ΔG_{exp} = −8.6 kcal/mol). On the other hand, QM-MM/GBSA outperformed (ΔG_{calc} = −9.5 ± 1.5 kcal/mol) both classical approaches, in agreement with previous findings.[59] QM-MM/GBSA success relies on using a DFTB Hamiltonian, which better describes the interactions between the active site residues and Me-α-L-Fucose, particularly the dispersion effects.

CONCLUSIONS

gmx_MMPBSA is a program written in Python aiming to perform end-state free energy calculations with GROMACS files. Our tool simplifies end-state free energy calculations with implicit solvent models while automating the input file preparation process to minimize user errors. Two additional tools, gmx_MMPBSA_test and gmx_MMPBSA_ana, have been included to enhance gmx_MMPBSA usability. gmx_MMPBSA_test has been designed for testing purposes and familiarizing the user with each supported type of calculation as well as all kinds of systems that can be studied with gmx_MMPBSA. On the other hand, gmx_MMPBSA_ana allows the user to visualize and analyze results from gmx_MMPBSA output files.

gmx_MMPBSA brings improvements in compatibility, versatility, scope of the analyses, and parallelization compared to existing programs. Our tool is compatible with all GROMACS versions, which has been a constant issue with programs with a similar purpose. gmx_MMPBSA offers access to various calculations with different methods, including binding free energy calculations, computational alanine scanning, entropy estimation, and binding free energy decomposition with different schemes. gmx_MMPBSA_ana is a dedicated analysis tool that provides the user access to all gmx_MMPBSA results. Outcomes from up to 12 different methods can be shown simultaneously for several systems. Last but not least, all calculations and analyses can be speeded up through parallelization with MPI.

Since its initial release in October 2020, gmx_MMPBSA has had tremendous acceptance within the GROMACS community (ca. >12 500 downloads and >1500 users accessing the documentation, as of August 2021), which has pushed us forward to create a robust toolkit. Thorough documentation is available with a detailed explanation for all the routines implemented in gmx_MMPBSA and various illustrating examples. The latest version (v1.4.3, 26/05/2021) of gmx_MMPBSA is distributed under the GNU General Public License (version 3) and available for download at https://github.com/Valdes-Tresanco-MS/gmx_MMPBSA. Documentation and tutorials are also available at https://valdes-tresanco-ms.github.io/gmx_MMPBSA.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.1c00645.

Details on systems preparation, MD simulation procedures, and energy calculations; topology generation process for AMBER and CHARMM force fields (Figure S1); time evolution of instantaneous per-residue

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energies throughout the MD simulation (Figure S2); and enthalpic (ΔH) and entropic ($-T\Delta S$) components for wild-type and mutant systems (Figure S3) (PDF)

AUTHOR INFORMATION

Corresponding Authors

Mario S. Valdés-Tresanco − *Faculty of Basic Sciences, University of Medellin, Medellin 050026, Colombia*; Email: mariosergiovaldes145@gmail.com

Ernesto Moreno − *Faculty of Basic Sciences, University of Medellin, Medellin 050026, Colombia*; orcid.org/0000-0002-2309-4826; Email: emoreno@udem.edu.co

Authors

Mario E. Valdés-Tresanco − *Centre for Molecular Simulations and Department of Biological Sciences, University of Calgary, Calgary, Alberta T2N 1N4, Canada*

Pedro A. Valiente − *Donnelly Centre for Cellular & Biomolecular Research, University of Toronto, Toronto, Ontario M5S 3E1, Canada; Center of Protein Studies, Faculty of Biology, University of Havana, 25 & J 10400 La Habana, Cuba*; orcid.org/0000-0002-4776-6017

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jctc.1c00645

Author Contributions

[§](M.S.V-T. and M.E.V-T.) These authors contributed equally.

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Notes

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ABBREVIATIONS

MM/PB(GB)SA, molecular mechanics/Poisson−Boltzmann (Generalized-Born) surface area; 3D-RISM, 3D-Reference Interaction Site Model; MD, molecular dynamics; SAS, solvent accessible surface area; NMODE, normal modes analysis; QH, quasi-harmonic; IE, interaction entropy; STP, single trajectory protocol; MTP, multiple trajectory protocol; QM/MM, quantum mechanical−molecular mechanical

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